



Conan O'Craft

Conan O'Brien tries to understand World of Warcraft. See what he has to say: <http://dgit.in/conancraft>



Windroid?

Intel trying to turn Android into Windows? Is the dominance of Android in the market enough to make Intel follow the masses? http://dgit.in/win_droid

Bazaar

Intel Core i7 4960X

Two years after the 3960X's introduction, its successor emerges!

The enthusiast segment rarely sees any action and the last time we saw a processor enter the segment was two years ago. For the investment they require, one would expect to ride the wave for a few years before having to go in for an upgrade. Also, the processors aren't cheap nor are the motherboards built for them. However, this isn't the deciding factor, it's the sheer ridiculous power that these machines output that make them objects of envy. That being said, the latest entrant into the enthusiast market seems to lack the excitement that usually accompanies a launch like this. Maybe Intel is doing the customers a favour by offering them a transitional processor to bide time till Haswell-E launches same time around next year. Which would explain the processor using the same platform as before. You will need to get a BIOS update before you can stick this processor in your computer as most old motherboards won't recognise the processor out of the box but those who've gone in for an Intel board will be disappointed to know that Intel has officially announced that they won't be enabling backwards compatibility with the IVB-E motherboard.

Like its predecessor the 4960X comes with an unlocked multiplier that scales up to 63x depending on the BCLK, so the multiplier can be varied while the BCLK is set to 100MHz, 125MHz and 166MHz. While the 3960X was set to 3.3GHz and then had a boost frequency of 3.9GHz the 4960X starts off at 3.6GHz and tops off at 4GHz under turbo. The Turbo barely seems different and the benchmarks pretty much reflect the same. The TDP is the same and it still features 40 PCIe lanes though we didn't have any problems like graphics cards running on PCIe 2.0 instead of PCIe 3.0 which is what happened with the 3960X, so native PCIe 3.0 support is guaranteed. What comes with this new iteration is that it's manufactured using a 22nm process. The maximum memory bandwidth has also increased by 8.5 GHz but that's something which is more dependent on the motherboard manufacturer who decides which SKU can support which frequency memory. The 3960X was obtained from a Xeon processor by knocking off two processor cores, however, the 4960X seems to have six cores natively.

For benchmarks we used the ASRock Extreme9 motherboard with 4 x 4GB ADATA memory kit running at 2133MHz. The monitor was an ASUS VK278 and the cooler was the stock model we had



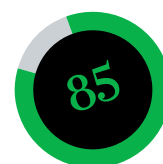
when the 3960X came in and for the graphics card we had a ZOTAC 780. The whole rig was powered by a Corsair RM650 power supply.

Performance wise, the 4960X has only managed to get a meagre 5-8% more compared to the 3960X. Video encoding benchmarks scores show that the 4960X was capable of getting 98.66 fps while the 3960X could only manage 90.59 fps. Gaming benchmarks as expected showed no difference at all, a few FPS here and there but all within the margin or error. Blender scene rendering test put the 4960X ahead since it managed to complete the scene in 21.16 seconds while the 3960X did the same scene in 25.51 seconds.

Like the 3960X even the 4960X was introduced with a closed loop liquid (propylene glycol) cooling system and it looks exactly the same as before, we didn't get a test system with the processor so we won't be able to give you the numbers on that. The motor is attached to the CPU block and the radiator has an in-built buffer which can be called a reservoir if we are to be picky. Overall, the device seems to be an Asetek OEM which is quite reliable.

Like we mentioned before, this seems like a transitional processor just to give enthusiasts an avenue to spend what they usually do to get the edge over everyone though we'd recommend sticking to your current enthusiast system for another year when the Haswell-E processors will be released along with a platform refresh bringing the X99 on to the scene. Did we mentioned that it's going to support DDR4 memory starting at 2133MHz? That will be a refreshing change.

Mithun Mobandas



Performance..... 95
Value for money..... 75

Specifications

Intel Core i7 4960X;
Ivy Bridge-E; Cores 6;
Hyperthreading support;
Multiplier unlocked
Frequency 3600 MHz; Turbo
Frequency 4000 MHz;
Clock Multiplier 36; Socket
LGA2011; L3 Cache 15 MB; 40
PCIe Lanes; Quad Channel
memory support; TDP 130W

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